

	Document Type:	FORM ISO 9001:2015	Document Code	RES-F08
	Document Title:	COMPLETED RESEARCH	Revision No.	00
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(1) Research Program/Project/Study Title: Anchovies <i>Engraulidae</i> Ice cream	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:			
Province:			
Region:			
(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
(7) Sector/Commodity SEAFOODS COMMODITY		(8) Discipline/Field of Specialization CULINARY	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Anchovies have many vitamins and minerals that provide major health benefits. They are best known as a source of omega-3 fatty acids, which promote brain and heart health. On the other hand, ice cream is a highly popular, palatable, nutritious, and commercially important dairy product, usually made from dairy products, such as milk and cream, and often combined with fruits or other ingredients and flavors. The researcher took advantage of the abundance in the supply of anchovies. Known for its health benefits, the researcher incorporated anchovies in ice cream to create a product that is highly nutritious without compromising palatability. The study is limited to the determination of the acceptability of the anchovies ice cream in terms of color, odor, taste, and mouthfeel. The pH of the product was also determined. There were 4 treatments: A (25% anchovies w/w), B (50% anchovies w/w), C (75% anchovies w/w) and D (100% anchovies w/w). Based on the data gathered there was no significant difference in the acceptability of the 4 treatments of anchovies ice cream in terms of its acceptability level in all sensory parameters as well as on its general acceptability. Overall all treatments had a mean score of “like slightly” in their acceptability score. In terms of pH, it was noted that the more anchovies added, the more acidic the product becomes. Although there was no significant difference, the researcher labeled treatment B as the most acceptable since in almost all parameters it is treatment B that has the highest acceptability score. The study showed that anchovies were acceptable raw material for desserts and varying amounts does not affect the acceptability.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Seafood ice cream, fish ice cream, seafood dessert, ice cream, anchovy

(9) Rationale:

Ice cream is a highly popular, palatable, nutritious, and commercially important dairy product. It is a valuable accompaniment to the normal diet for all age groups, especially for children. Ice cream is a sweetened frozen food typically eaten as a snack or dessert. It is usually made from dairy products, such as milk and cream, and often combined with fruits or other ingredients and flavors. One of the most common fish you'll find in the Philippines is anchovies fish. Just as the fish is unique, it also has a distinct taste that makes it a favorite recipe for the Filipino family. Ice cream is always in the vocabulary of every individual. In this study, the researcher aims to produce an ice cream out of anchovies fish. They want to know if this ice cream will be popular with everyone, especially kids who love ice cream. Also, they want to know if this ice cream will be known for its different and unique characteristics.

The dairy product prepared by pasteurization, homogenization, aeration and freezing that has been maintained at uniform consistency is called ice cream. It is a cheap, healthy, nutritious, and palatable product. The composition of ice cream is comprised of sugar, fat, emulsifiers, stabilizers, water, egg and egg products, corn syrup, dextrose, and flavors. It is a three-phase network consisting of on-air, solid, and liquid in the final product. The liquid phase contains ice crystals in embedded form and air cells in dispersed form. Milk proteins, soluble and insoluble salts, fat particles, stabilizers, and sugars are also present in a liquid phase. So it is a very complex physiochemical food system. Ice cream is categorized into dairy frozen desserts in which frozen confections, water ice, sherbet, frozen custard, pareve-like products, ice milk, and mellorine are included. Anchovies have many vitamins and minerals that provide major health benefits. They are best known as a source of [omega-3 fatty acids](#), which promote brain and heart health. Anchovies also have selenium, which, if eaten regularly, may reduce the risk of some types of cancer.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

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Materials:

Anchovies Fish
 Roselle leaves (labog)
 Ginger
 Coconut cream
 All-purpose cream
 Condense Milk

Experimental Treatment

Ingredients	Treatment A	Treatment B	Treatment C	Treatment D
Anchovies fish	25%	50%	75%	100%

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

RESULTS AND DISCUSSION

Sensory Evaluation

Scoring of Variables

The criteria for evaluation as to sensory characteristics of the product was 5-point hedonic scale in terms of color, odor, mouth feel and taste. The rating for general acceptability in 9-point hedonic scale was: 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).

The score on each item was interpreted based on the result of the computation as to color, odor, texture, taste and general acceptability using the following scoring interval and verbal interpretation.

For Color

Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
5	White	4.2-5.0
4	Creamy	3.4-4.9
3	Light Gray	2.6-3.39
2	Dirty White	1.8-2.59
1	Beige	1.0-1.79

For Odor

Scoring	Verbal interpretation/ Response Categories	Scoring Interval
5	No fishy odor	4.2-5.0
4	Slightly fishy smell	3.4-4.9
3	Moderately fishy smell	2.6-3.39

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2	Very much fishy smell	1.8-2.59
1	Extremely fishy smell	1.0-1.79
For Mouth Fell		
Scoring	Verbal Interpretation/ Response Categories	Scoring Interval
5	Extremely smooth	4.2-5.0
4	Very much smooth	3.4-4.9
3	Moderately smooth	2.6-3.39
2	Slightly smooth	1.8-2.59
1	Slightly rough	1.0-1.79
For Taste		
Scoring	Verbal interpretation/ Response Categories	Scoring Interval
5	Extremely delicious	4.2-5.0
4	Very much delicious	3.4-4.9
3	Much delicious	2.6-3.39
2	Moderately delicious	1.8-2.59
1	Slightly delicious	1.0-1.79
For Acceptability		
Scoring	Score Interval	Verbal interpretation/Response Categories
9	8.12 – 9.0	Like Extremely
8	7.23 – 8.11	Like Very Much
7	6.34 – 7.22	Like Moderately
6	5.45 – 6.33	Like Slightly
5	4.56 – 5.44	Neither Like nor Dislike
4	3.67 – 4.55	Dislike Slightly
3	2.78 – 3.66	Dislike Moderately
2	1.89 – 2.77	Dislike Very Much
1	1.00 – 1.88	Dislike Extremely
Table 1.0 Descriptive statistics of the sensory characteristics in terms of color, taste, mouthfeel, and odor using 5-point hedonic scale.		

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Variable	Treatment	Mean	Std. Deviation	N
Color	1	3.80	1.243	30
	2	3.77	1.278	30
	3	3.90	1.185	30
	4	3.83	1.117	30
	Total	3.83	1.193	120
Taste	1	3.60	.894	30
	2	3.53	.937	30
	3	3.57	.858	30
	4	3.33	.884	30
	Total	3.51	.889	120
Mouthfeel	1	3.57	.858	30
	2	3.73	.691	30
	3	3.50	.861	30
	4	3.47	.681	30
	Total	3.57	.775	120
Odor	1	3.90	.845	30
	2	4.07	.785	30
	3	3.67	.922	30
	4	3.97	.850	30
	Total	3.90	.854	120
Total	1	3.72	.972	120
	2	3.77	.957	120
	3	3.66	.966	120
	4	3.65	.923	120
	Total	3.70	.953	480

*Legend: Refers to the Scoring of variables

Sensory characteristics limited to color, taste, mouthfeel, and odor. Five-point hedonic scale was used with 10 trained panelists with three trials.

In terms of color and odor, all the treatments, when rounded up, has a mean score of 4 which is equivalent to Creamy(color) and Slightly fishy smell (odor). With taste and mouthfeel, treatment A, B and C has a mean score of 4 while treatment D has a mean score of 3 is equivalent to much delicious(taste) and moderately smooth(mouth feel). None of the treatments garnered a score lower than 3 which makes the sensory characteristics ideal.

Table 2.0 One-way ANOVA results of the sensory characteristics in terms of color, taste, mouthfeel, odor.

	Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Contrast	13.217	3	4.406	4.912	.002	.031
Error	416.133	464	.897			

*The F tests the effect of the Variable. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.

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Using statistical analysis, one-way ANOVA, results showed that there is a significant difference with the sensory characteristics in terms of parameter (color, odor, mouthfeel, and taste).

Table 3.0 Post-hoc test for the sensory characteristics in terms of color, odor, mouthfeel, and taste.

	(I) Variable	(J) Variable	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Tukey HSD	Color	Taste	.32*	.122	.048	.00	.63
		Mouthfeel	.26	.122	.150	-.06	.57
		Odor	-.08	.122	.928	-.39	.24
	Taste	Color	-.32*	.122	.048	-.63	.00
		Mouthfeel	-.06	.122	.964	-.37	.26
		Odor	-.39*	.122	.008	-.71	-.08
	Mouthfeel	Color	-.26	.122	.150	-.57	.06
		Taste	.06	.122	.964	-.26	.37
		Odor	-.33*	.122	.034	-.65	-.02
	Odor	Color	.08	.122	.928	-.24	.39
		Taste	.39*	.122	.008	.08	.71
		Mouthfeel	.33*	.122	.034	.02	.65

Based on observed means.

The error term is Mean Square(Error) = .897.

*. The mean difference is significant at the .05 level.

Since there was a significant difference in the parameters of the sensory characteristics, a post-hoc test (Tukey test) was conducted. It could be seen in table 3.0, that there is a significant difference in the score of color vs taste, taste vs odor, mouthfeel vs odor. This simply means that the mentioned parameter has an interaction which impacted their judgement and resulted to significant difference with their scoring.

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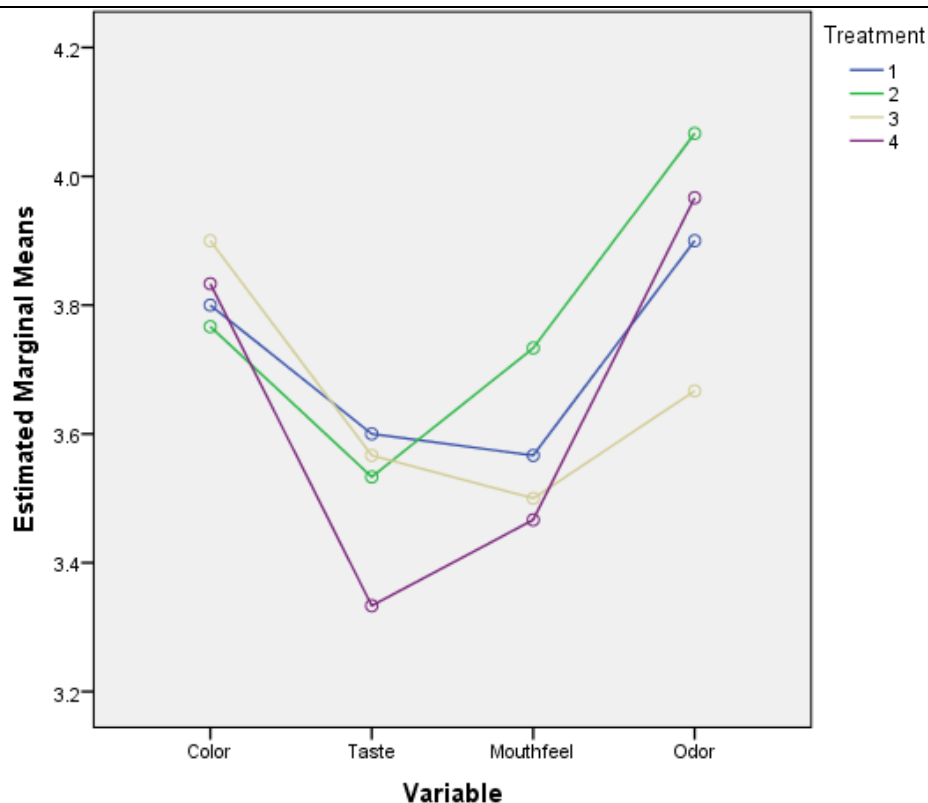


Figure 1.0 The mean of the results of the sensory characteristics of the 4 treatments.

As presented on the graph above, the most acceptable treatment is treatment B. Since there is no significant difference statistically, the researcher decided to adopt the highest scoring based on the summary of the individual parameters.

Table 4.0 Descriptive statistics of the general acceptability of the 4 treatments using 9-point hedonic scale.

Treatment	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
Treatment 1	5.767	.352	5.069	6.464
Treatment 2	5.700	.352	5.003	6.397
Treatment 3	5.667	.352	4.969	6.364
Treatment 4	5.633	.352	4.936	6.331

*Legend: 1= Dislike Extremely, 2= Dislike Very Much, 3= Dislike Moderately, 4= Dislike Slightly, 5= Neither Like nor Dislike, 6= Like Slightly, 7= Like Moderately, 8= Like Very Much, 9= Like Extremely

Shown in Table 4.0 is the descriptive statistics of the general acceptability of the 4 treatments using a 9-point hedonic scale. Based on the data gathered, the mean score of all the treatments is 6, which is like slightly. The same equivalent score is observed with the specific sensory parameters in Table 1.0.

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Table 5.0 One way ANOVA results of the general acceptability of the 4 treatments.

	Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
Contrast	1.217	3	.406	.452	.716	.003
Error	416.133	464	.897			

The F tests the effect of Treatment. This test is based on the linearly independent pairwise comparisons among the estimated marginal means.

Using one-way ANOVA, results showed that there is no significant difference in the acceptability level of the 4 treatments. Thus, making treatment B a valid choice as the most acceptable treatment.

pH

Table 5.0 Descriptive statistics of the pH value of the 4 treatments.

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Min	Max
					Lower Bound	Upper Bound		
Treatment A	3	8.2467	.59501	.34353	6.7686	9.7247	7.59	8.75
Treatment B	3	7.2000	.13528	.07810	6.8640	7.5360	7.07	7.34
Treatment C	3	6.9167	.05686	.03283	6.7754	7.0579	6.87	6.98
Treatment D	3	6.7067	.04509	.02603	6.5947	6.8187	6.66	6.75
Total	12	7.2675	.67136	.19381	6.8409	7.6941	6.66	8.75

The mean pH value of the 4 treatments is shown in table 5.0. Based on the data gathered, there is a decreasing trend as the anchovies are increased in the formulation. Treatment A (pH 8.25) is found to be basic while treatment B (pH 7.20) is neutral while the rest of the treatments, C (pH 6.92) and D (pH 6.71), are acidic. pH has a direct effect on the taste. As the sample increases in acidity, more sourness is detected while basic taste reflects bitterness (Da Conceicao Neta, E. R., Johanningsmeier, S. D., Drake, M. A., & McFeeters, R. F. ,2007).

Table 6.0 One-way ANOVA results of the pH value of the 4 treatments.

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	4.203	3	1.401	14.840	.001
Within Groups	.755	8	.094		
Total	4.958	11			

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The pH value of the 4 treatments were statistically tested using one way ANOVA. Results showed that there is a significant difference with the pH value of the 4 treatments.

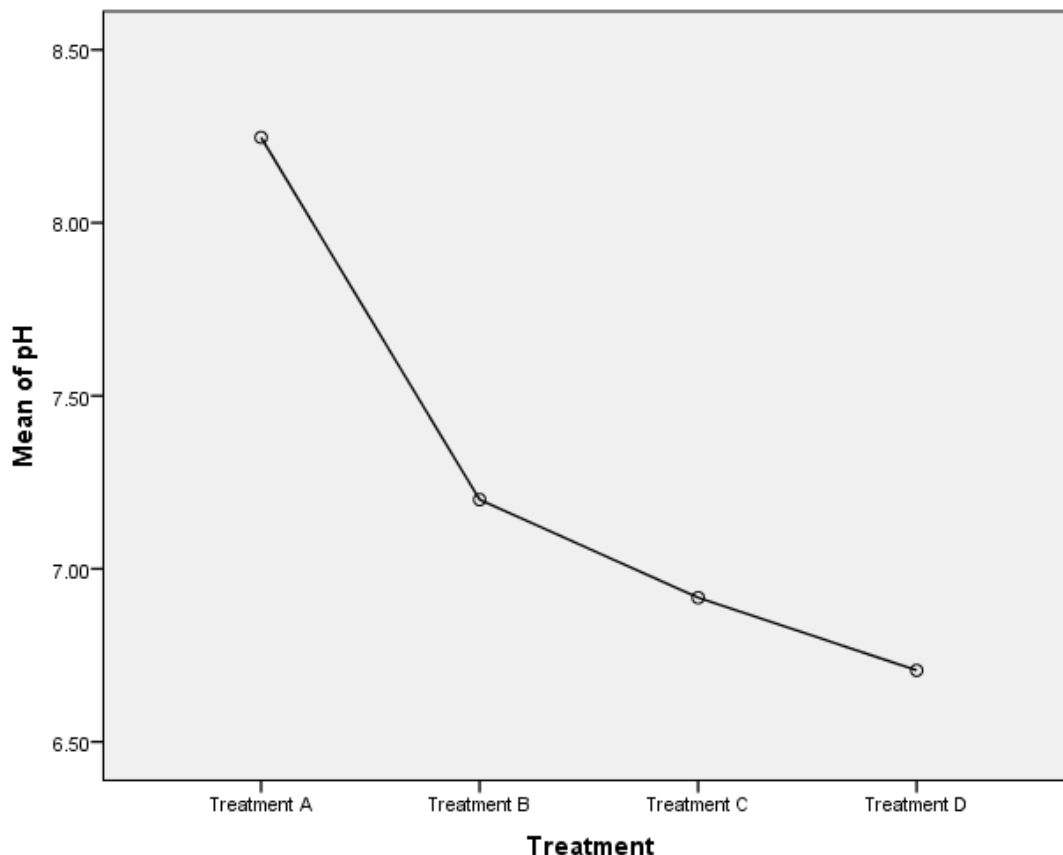
Table 7.0 Post hoc test for the pH value of the 4 treatments.

	(I) Treatment	(J) Treatment	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Tukey HSD	Treatment A	Treatment B	1.04667*	.25087	.013	.2433	1.8500
		Treatment C	1.33000*	.25087	.003	.5266	2.1334
		Treatment D	1.54000*	.25087	.001	.7366	2.3434
	Treatment B	Treatment A	-1.04667*	.25087	.013	-1.8500	-.2433
		Treatment C	.28333	.25087	.683	-.5200	1.0867
		Treatment D	.49333	.25087	.276	-.3100	1.2967
	Treatment C	Treatment A	-1.33000*	.25087	.003	-2.1334	-.5266
		Treatment B	-.28333	.25087	.683	-1.0867	.5200
		Treatment D	.21000	.25087	.836	-.5934	1.0134
	Treatment D	Treatment A	-1.54000*	.25087	.001	-2.3434	-.7366
		Treatment B	-.49333	.25087	.276	-1.2967	.3100
		Treatment C	-.21000	.25087	.836	-1.0134	.5934

*. The mean difference is significant at the 0.05 level.

To determine where the significant difference occurred, a post hoc test was conducted, specifically tukey test. Based on the results, it showed that the pH value of treatment A has a significant difference with the rest of the treatments.

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Figur 2.0 Mean of the pH value of the 4 treatments.

Visually illustrating the results of the post hoc test, figure 2.0 reflects the graph of the mean pH value of the 4 treatments. A decreasing trend could also be observed. As the researcher increased the amount of anchovy in the formulation of the ice cream the pH value decreased. The decrease in pH was brought by the presence of anchovy which is acidic in nature, where the fresh anchovy's pH ranges from 6.32 to 6.50 (Kayim, M. and Can,E., 2010).

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

CONCLUSION AND RECOMMENDATIONS

The utilization of anchovies in the production of ice cream is acceptable with a general rating of slightly liked. All sensory parameters are color, odor, taste, mouthfeel are acceptable with a rating of slightly liked as well. This product is new to the consumers which created a new flavor sensation. This does not just give a healthy option but also provides added value to the anchovies which are known for their health benefits and abundance at the site of implementation.

In terms of pH value, the treatment was all within the standard. The increase in anchovies (w/v) increased its acidity level but it did not create any adverse effect on the taste since the results of the sensory evaluation showed a rating of slightly liked with no significant difference.

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(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Anchovies Ice Cream

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

Production of Anchovies Ice Cream

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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